Ruturaj Mane

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Github link Portfolio link

PROFESSIONAL SUMMARY

Data Scientist with strong analytical skills and hands-on experience in machine learning and statistical modeling. I work with complex datasets to find meaningful patterns, create reliable predictive machine learning models, and improve their evaluation through model optimization and hyperparameter tuning. I design and improve scalable machine learning solutions that turn data insights into strong, effective systems ready for real-world use.

WORK EXPERIENCE

DATA ANALYST INTERN, *SmartBridge* 07/2025 – 10/2025

- Extracted, cleaned, and transformed large datasets using SQL (PostgreSQL) and Python (Pandas, NumPy) to build analysis-ready datasets, improving data accuracy and reliability by 18%.
- Conducted exploratory data analysis, anomaly detection, and performance analytics to identify trends, risks, and data-quality gaps, leading to a 35% improvement in reporting effectiveness.
- Designed KPI-driven Tableau dashboards in collaboration with cross-functional teams, ensuring data validation, consistency, and audit-ready analytics outputs.

SKILLS SUMMARY

Technical skills
Python, SQL, Excel, Tableau, Power BI, Machine learning

Analytical & Data Skills
Exploratory Data Analysis (EDA), Statistical Analysis, KPI Design, Dashboard Development, Data Cleaning&Modeling

Libraries & Frameworks
Pandas, NumPy, Scikit-learn, Seaborn, Matplotlib

Machine Learning & AI Skills
Supervised & Unsupervised Learning, Regression & Classification, Feature Engineering, Model Training and Testing, Model Evaluation

Projects

NEET 2024 Result Analysis & Prediction , 08/2025

| *Python(pandas, numpy, scikit-learn), Machine learning, Tableau*

- Analyzed NEET 2024 data from over 1M+ records to find performance patterns and regional trends across 29 states. This generated insights that are useful for educational decision-making.
- Processed large datasets using Pandas and NumPy. I applied Scikit-learn models, including K-Means and Random Forest, for trend prediction and segmentation. This improved overall analytical accuracy by 35%.
- Integrated predictive insights into Tableau stories to compare actual and predicted outcomes. This allowed for clear reporting, risk identification, and data-driven interpretation.

Smart Agriculture Analytics & Recommendation System , 09/2025

| *Python(pandas, duckDB), Machine learning, Tableau*

- Developed a data-driven agriculture analytics system, This system combined multiple datasets to study crop yield, production patterns, and suggestions for better farming efficiency.
- Built an Analytical Base Table (ABT), conducted exploratory data analysis (EDA), and standardized three datasets. This process established a reliable data flow and modeling structure for recognizing crop patterns using machine learning.
- Provided real-time insights using Tableau and a Streamlit application. This achieved 99% model accuracy, allowed for KPI monitoring, and supported data-driven agricultural planning for stakeholders.

Deep Learning–Based Road Segmentation from Satellite Imagery , 01/2026

| *Python (PyTorch), U-Net, Deep Learning, Computer Vision, Streamlit*

- Developed an automated road extraction system using binary semantic segmentation on over 1,100 high-resolution satellite image-mask pairs. Implemented a 4-level encoder-decoder U-Net architecture with PyTorch.
- Built a structured preprocessing and training pipeline (RGB normalization, 256×256 resizing, Adam optimizer with BCE + Dice loss) and achieved ~0.74 Dice coefficient and ~0.68 IoU on unseen test data.
- Deployed the trained model as an interactive Streamlit web app. This allows for real-time CPU inference, taking less than 300 ms per image. Users can upload images, view masks, and download segmentation outputs.

EDUCATION

D Y Patil School Of Engineering And Technology Pune,India
Bachelor of Technology - Computer Engineering (SGPA: 9.15) 06/2022 -08/2026

RELEVANT CERTIFICATES

- Google Data Analytics (Google)
- Data Fundamentals (IBM)